

News Application (NewsSnap)

1. Introduction

NewsSnap is a modern web-based news aggregation platform designed to provide users with short and concise news summaries. In today's fast-paced world, people often struggle to stay updated with current events because traditional news platforms provide lengthy articles that require significant time to read.

NewsSnap addresses this problem by delivering news in a summarized format of **60–100 words**, allowing users to quickly grasp important updates. The platform organizes news into various categories such as **Technology, Sports, Business, Entertainment, Politics, and more**, enabling users to easily navigate topics of interest.

Built using the **MERN stack (MongoDB, Express.js, React.js, Node.js)**, the application offers a responsive interface, efficient backend processing, and structured storage of news data.

2. Description

NewsSnap is a web-based news aggregation application designed to deliver short, concise, and easily readable news summaries across multiple categories such as Technology, Sports, Business, Entertainment, Politics, and more.

Inspired by platforms like **Inshorts**, the system focuses on presenting important news in a brief format (approximately **60–100 words**) so users can quickly stay informed without spending much time reading lengthy articles.

The platform collects and organizes news from multiple trusted sources and converts them into summarized formats highlighting the key points of each story. Users can browse different categories, view trending news, and scroll through summaries to stay updated with the latest events.

The application provides a **clean and user-friendly interface** optimized for both desktop and mobile devices, ensuring a smooth and convenient news-reading experience.

3. Problem Statement

Many people find it difficult to keep up with current events due to lengthy news articles and limited time. Traditional news platforms often provide excessive information, making it harder to quickly understand key updates.

A concise news delivery system is needed to simplify news consumption and help users stay informed efficiently.

The objectives of the system are:

1. To develop a news aggregation platform that provides short news summaries.
2. To help users quickly stay updated with current events.
3. To categorize news into sections such as Technology, Sports, Business, and Entertainment.
4. To present news in a concise format (60–100 words).
5. To provide a clean and user-friendly interface for reading news.
6. To collect and organize news from multiple trusted sources.
7. To improve accessibility to important news updates in less time.

4. Scenario

Consider a **busy professional who wants to stay updated with daily news but has limited time**.

Instead of reading long articles from different websites, the user opens **NewsSnap** on their phone or laptop.

The user can quickly browse categories such as:

- Technology
- Sports
- Business
- Entertainment

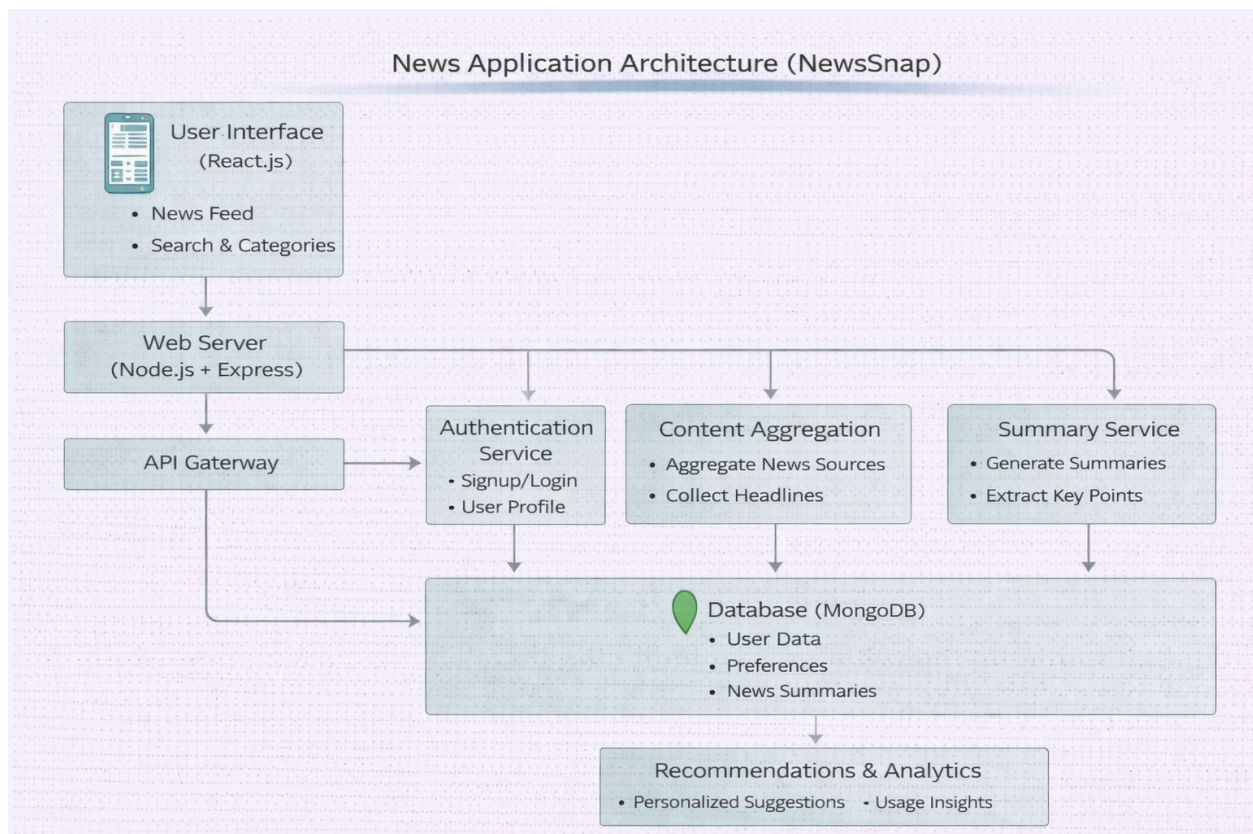
- Politics

Each news story is presented in a **short summary format**. The user can scroll through multiple stories in a few minutes and stay informed about the latest events.

If the user wants more details, they can click on the **original source link** to read the full article.

This approach allows users to **consume news quickly and efficiently**.

5. Architecture



The **NewsSnap** application follows a **MERN stack architecture** consisting of three primary layers.

Frontend Layer

The frontend is developed using **React.js**, providing a dynamic and responsive interface for browsing news articles.

Backend Layer

The backend is built using **Node.js and Express.js**, which handles API requests, processes news data, and manages communication between the frontend and database.

Database Layer

MongoDB is used to store news articles, categories, sources, and user interactions.

External API Integration

The system may use **News APIs** to fetch real-time news updates from trusted sources.

The frontend communicates with the backend through **REST APIs** to fetch and display summarized news.

6. Project Flow (System Flow)

Step 1: Fetch News

The backend fetches news articles from external **News APIs** or trusted sources.

Step 2: Data Processing

The system processes and summarizes the news content.

Step 3: Store News

Summarized news articles are stored in the **MongoDB database**.

Step 4: Categorization

News articles are categorized into sections such as Technology, Sports, and Business.

Step 5: User Access

Users open the application and browse news by category.

Step 6: Display News

The frontend displays short summaries of news articles.

Step 7: Read More

Users can click a link to read the full article from the original source.

7. User Flow

Opening the Application

User → Open NewsSnap → Home Page

Browsing News

User → Select Category → View News Summaries

Reading News

User → Click News Card → View Summary → Optional "Read More"

Scrolling Through News

User → Scroll Through Cards → View Multiple News Stories Quickly

8. Pre-Requisites

Frontend (React)

- React Components
- React Hooks
- React Router
- Axios for API requests
- Responsive design using CSS/TailwindCSS

Backend (Node.js & Express)

- Node.js fundamentals
- Express.js routing
- REST API development
- Middleware usage

Database (MongoDB)

- MongoDB document model
- CRUD operations
- Mongoose ODM
- Schema design

API Integration

- Working with external APIs
- JSON data handling

9. Required Technologies

Frontend

- React.js
- TailwindCSS / CSS
- Axios

Backend

- Node.js

- Express.js

Database

- MongoDB
- Mongoose

APIs

- News API (for fetching news)

Development Tools

- Git
 - GitHub
 - VS Code
 - Postman / Thunder Client
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10. Suggested Database Collections

Users (Optional)

User

- _id
- username
- email
- password
- preferences
- createdAt

News

News

- _id
- title

- summary
- category
- source
- imageUrl
- publishedAt

Categories

Category

- _id
- categoryName

Sources

Source

- _id
 - sourceName
 - sourceUrl
-

11. Key Features

News Summaries

- Short news format (60–100 words)
- Quick reading experience

News Categories

- Technology
- Sports
- Business
- Entertainment
- Politics

News Aggregation

- Collect news from multiple trusted sources

Responsive Interface

- Mobile-friendly design
- Clean and simple UI

Read More Option

- Link to original article source
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12. Optional Advanced Features

To enhance the platform further, the following features can be implemented:

- **AI-based news summarization**
 - **Personalized news recommendations**
 - **Trending news detection**
 - **Bookmark or save news feature**
 - **Dark mode interface**
 - **Push notifications for breaking news**
 - **User preference-based news feed**
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13. Learning Outcomes

By completing this project, developers will learn:

- Full-stack **MERN development**

- **API integration** with external services
- Data processing and **content summarization**
- Designing **news aggregation platforms**
- Building **responsive and user-friendly interfaces**
- Managing structured data using **MongoDB**
- Developing scalable web applications